

MA1513 Exam (Exemplify)

AY2023/2024 Semester I

MCQ/FIB (15 questions; 3 marks each)

1. A 3 by 4 homogeneous linear system has general solution $\begin{cases} x_1 = s - 2t \\ x_2 = t - 2s \\ x_3 = s \\ x_4 = t \end{cases}$. The reduced row echelon form of the augmented matrix of the system is given by $\left(\begin{array}{cccc|c} 1 & 0 & a & b & 0 \\ 0 & 1 & c & d & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$

where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$, $d = \underline{\hspace{1cm}}$ (FIB)

Ans: $a = -1$, $b = 2$, $c = 2$, $d = -1$

Eliminate s and t in the general solution:
Substitute s by x_3 and t by x_4 into x_1 and x_2 :
 $x_1 = x_3 - 2(x_4) \rightarrow x_1 - x_3 + 2(x_4) = 0$.
So first row of REF is $(1 \ 0 \ -1 \ 2 \ 0)$
 $x_2 = x_4 - 2(x_3) \rightarrow x_2 + 2(x_3) - x_4 = 0$.
So second row of REF is $(0 \ 1 \ 2 \ -1 \ 0)$

2. Suppose $A = (c_1 \ c_2 \ c_3)$ is a 3×3 singular matrix, where c_1, c_2, c_3 are the three columns of A . Which of the following statements are always true? (Pick all correct options)

- a. c_1 is a linear combination of c_2 and c_3 . It is possible that c_2 and c_3 are scalar multiple of each other, while c_1 is independent of c_2 and c_3 .
b. $\{c_1, c_2, c_3\}$ forms a basis for $\mathbb{R}^3$. If A is singular, c_1, c_2, c_3 must be linearly dependent.
c. There is a non-pivot column in any row echelon form of A . This is true for REF of singular matrix
d. The nullity of A is not 0. Since rank of $A < 3$, then nullity of A must be > 0
e. The linear system $Ax = b$ has infinitely many solutions for any 3-vector b . It is possible that for some b , $Ax = b$ has no solution.

3. Which of the following have the same value as the 3×3 determinant $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ d & e & f \end{vmatrix}$ for any values of a, b, c, d, e and f ? (Pick all correct options)

- a. $\begin{vmatrix} b & c \\ e & f \end{vmatrix} + \begin{vmatrix} a & c \\ d & f \end{vmatrix} + \begin{vmatrix} a & b \\ d & e \end{vmatrix}$ middle term should have negative sign
b. $a \begin{vmatrix} 1 & 1 \\ e & f \end{vmatrix} - b \begin{vmatrix} 1 & 1 \\ d & f \end{vmatrix} + c \begin{vmatrix} 1 & 1 \\ d & e \end{vmatrix}$ The signs of all three terms should be $-, +, -$
c. $d \begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix} - e \begin{vmatrix} 1 & 1 \\ a & c \end{vmatrix} + f \begin{vmatrix} 1 & 1 \\ a & b \end{vmatrix}$ Interchange row 1 and 3, then interchange row 2 and 3. Apply expansion to get option c.
d. $\begin{vmatrix} a & b \\ d & e \end{vmatrix} + \begin{vmatrix} b & c \\ e & f \end{vmatrix} + \begin{vmatrix} c & a \\ f & d \end{vmatrix}$ Note that the third term is the negative of $\det[a \ c \ f]$
e. $\begin{vmatrix} b & c \\ e & f \end{vmatrix} - a \begin{vmatrix} 1 & 1 \\ e & f \end{vmatrix} + d \begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix}$ Taking the transpose of the given matrix, and apply expansion to get option e.

4. The vector $(3, 6, 9)$ belongs to the row space of which of the following matrices? (Pick all correct answers.)

- a. $\begin{pmatrix} 0 & 0 & 3 \\ 0 & 6 & 0 \\ 9 & 0 & 0 \end{pmatrix}$ $(3, 6, 9) = 3(0, 0, 3) + 1(0, 6, 0) + 1/3(9, 0, 0)$, which is a linear combination of the rows of this matrix.
b. $\begin{pmatrix} 3 & 3 & 3 \\ 6 & 6 & 6 \\ 9 & 9 & 9 \end{pmatrix}$ All three rows are scalar multiples of $(1, 1, 1)$. Any linear combinations of the rows are also scalar multiples of $(1, 1, 1)$. So $(3, 6, 9)$ does not belong to the row space of this matrix.
c. $\begin{pmatrix} 3 & 6 & 0 \\ 0 & 6 & 9 \end{pmatrix}$ Check directly that $(3, 6, 9)$ is not a linear combination of $(3, 6, 0)$ and $(0, 6, 9)$
d. $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ $(3, 6, 9) = 3(1, 2, 3) + 0(1, 1, 1) + (0, 0, 0)$ which is a linear combination of the rows of this matrix.

$$e. \begin{pmatrix} 3 & 3 & 3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \\ 6 & 3 & 0 \end{pmatrix}$$

$(3, 6, 9) = (3, 3, 3) + (0, 3, 3) + (0, 0, 3) + 0(6, 3, 0)$
which is a linear combination of the rows of this matrix.

5. Which of the following sets of vectors are bases for the subspace $\text{span}\{(1, 2, 1, 1), (1, 0, 2, -1), (1, 4, 0, 3), (-1, 0, -2, 1)\}$ of $\mathbb{R}^4$? (Pick all correct answers)
- a. $\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)\}$
- b. $\{(1, 2, 1, 1), (1, 0, 2, -1), (1, 4, 0, 3)\}$
- c. $\{(1, 0, 2, -1), (1, 4, 0, 3)\}$
- d. $\{(1, 2, 1, 1), (-1, 0, -2, 1)\}$
- e. $\{(1, 0, 2, 0), (0, 1, -1, -1)\}$
- First check that 2 of the 4 vectors in the spanning set are redundant (for example, stack the 4 vectors as columns of a matrix, and check its REF has only 2 pivot columns. So basis for the subspace must have exactly 2 vectors. So options a and b are out.
- Vectors in both options c and d come from the spanning set and are linearly independent. So they can be bases for the subspace.
- Check these two vectors are not linear combinations of the vectors in the spanning set, so they cannot be basis for the subspace.
6. Which of the following vectors will have the projection p on the plane with equation $x - 2y + z = 0$ where $p = (-1, 1, 3)$. (Pick all correct answers)
- a. $(1, -2, 1)$
- b. $(3, 0, 1)$
- c. $(0, -1, 4)$
- d. $(-2, 3, 2)$
- e. $(1, 0, -1)$
- The correct answer must be a vector v such that $p - v$ is orthogonal (perpendicular) to the plane. i.e. $p - v$ is scalar multiple of $(1, -2, 1)$ (this come from the coefficient of the equation of the plane).
- By going through the 5 options, we check that vectors in options c and d satisfy the above condition.
7. Suppose the regular x, y and z-axes in the XYZ space is rotated 45° clockwise about the z-axis (when view from top). The coordinate vector of $(3\sqrt{2}, \sqrt{2}, 3)$ relative to the new axes is given by (a, b, c) where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)
- The unit vectors along the three new axes are $(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0)$, $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)$, $(0, 0, 1)$.
- (Hint: The answers are whole numbers with no square root involved)
- Ans: $a = 2, b = 4, c = 3$
- Solve $(3\sqrt{2}, \sqrt{2}, 3) = a(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0) + b(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0) + c(0, 0, 1)$ to get $a = 2, b = 4, c = 3$
8. Suppose we are given the reduced row echelon form R of an unknown matrix A. Which of the following conditions allow us to determine the column space of A?
- a. R is full rank. R can be full rank with zero rows. This will not allow us to determine the column space
- b. R does not have any zero row. In this case, column space will be $\mathbb{R}^n$ where n is the number of rows of R.
- c. R has no non-pivot column. R can have no non-pivot column but with zero rows.
- d. Number of rows and columns of R are the same. It is possible for the RREF of a square matrix to have zero rows
- e. It is not possible to determine the column space of A without knowing A.
9. (Linear transformation) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation with standard matrix A. Suppose the first two columns of A are $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. Knowing the image of which of the following vectors will allow us to completely determine T? (Pick all correct answers)
- a. $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$
- b. $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$
- c. $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$
- d. $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$
- Since the first two columns of A are images of $[1; 0; 0]$ and $[0; 1; 0]$ respectively, in we need to know the image of a third vector v which is linearly independent of $[1; 0; 0]$ and $[0; 1; 0]$. In other words, the correct options should be a vector that are not linear combinations of $[1; 0; 0]$ and $[0; 1; 0]$ in order that the three vectors form a basis for $\mathbb{R}^3$.

e. Any vector except the linear combinations of $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

10. Given A is a 4x4 matrix and u, v, w, x are linearly independent vectors in $\mathbb{R}^4$. Suppose $Au = v$, $Av = u$, $Aw = 0$ and $Ax = x$.

Which of the following statements are correct? (Pick all correct answers)

- a. A has 3 distinct eigenvalues 0, 1, -1 are the three distinct eigenvalues
 b. A has only 2 linearly independent eigenvectors
 c. A has an eigenspace with dimension 2 The eigenspace for eigenvalue 1 is of dimension 2.
 d. A is not diagonalisable A is 4x4 with 4 lin.indep. eigenvectors
 e. A is singular Since 0 is an eigenvalue, A is singular

$Aw = 0 \rightarrow w$ is eigenvector with eigenvalue 0
 $Ax = x \rightarrow x$ is eigenvector with eigenvalue 1
 $A(u+v) = v+u \rightarrow u+v$ is eigenvector with eigenvalue 1
 $A(u-v) = v-u = -(u-v) \rightarrow u-v$ is eigenvector with eigenvalue -1
 There are 4 linearly independent eigenvectors w, x, $u+v$, $u-v$

11. In a certain town, 25% of the married men get divorced each year and 20% of the single men get married each year. There are 6000 married men and 3000 single men. Assuming that the total population of men remains constant over the years, there will be M married men and S single men in the long run, where $M = \underline{\hspace{1cm}}$, $S = \underline{\hspace{1cm}}$ (FIB).

Ans: $M = 4000$, $S = 5000$

The stage matrix is $A = \begin{bmatrix} 0.75 & 0.2 \\ 0.25 & 0.8 \end{bmatrix}$
 In the long run $A^n = \begin{bmatrix} -1 & 4/5 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 4/5 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 4/9 & 4/9 \\ 5/9 & 5/9 \end{bmatrix}$
 Population in long run: $\begin{bmatrix} 4/9 & 4/9 \\ 5/9 & 5/9 \end{bmatrix} \begin{bmatrix} 6000 \\ 3000 \end{bmatrix} = \begin{bmatrix} 4000 \\ 5000 \end{bmatrix}$

12. The general solution of a 3 by 3 SDE is given by

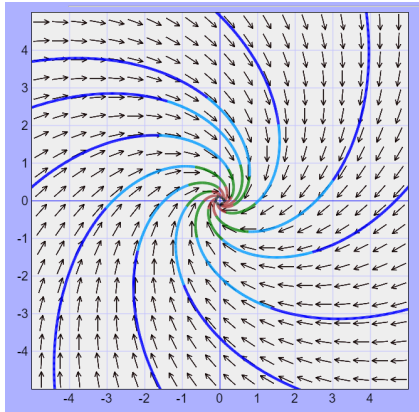
$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} e^{-7t} + c_2 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} e^{3t} + c_3 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} e^t. \text{ If an initial condition is given by } \begin{pmatrix} y_1(0) \\ y_2(0) \\ y_3(0) \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}.$$

Then $c_1 = \underline{\hspace{1cm}}$, $c_2 = \underline{\hspace{1cm}}$, $c_3 = \underline{\hspace{1cm}}$ (FIB)

Ans: $c_1 = 1$, $c_2 = -1$, $c_3 = 2$

Solving the system $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$ gives $\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

13. The phase portrait of a 2 by 2 SDE $y' = Ay$ is given below.



This is a spiral point that is stable and pass through the test point $[1; 0]$ with tangent vector pointing downward.

Which of the following is the underlying matrix A of the SDE?

- a. $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ eigenvalue $1+i$, $1-i$; unstable
 b. $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ eigenvalue $1+i$, $1-i$; unstable
 c. $\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$ eigenvalue $-1+i$, $-1-i$; stable. tangent vector at test point is $[-1; -1]$, pointing downward
 d. $\begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}$ eigenvalue $-1+i$, $-1-i$; stable. tangent vector at test point is $[-1; 1]$, pointing upward
 e. None of the above

14. The 3rd order ODE given by $2y''' + 6y'' - 4y' - 2y = 0$ can be converted to a 3 by 3 SDE given by

$$\begin{pmatrix} y_1' \\ y_2' \\ y_3' \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$y_3' = y''' = 1/2 (2y + 4y' - 6y'') = y_1 + 2y_2 - 3y_3$$

where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)

Ans: $a = 1$, $b = 2$, $c = -3$

15. Given the equilibrium point of a 2 by 2 SDE is a star node.

Which of the following statements are true? (Pick all correct answers)

a. We can find infinitely many linearly independent eigenvectors for the underlying matrix. There are infinitely many eigenvectors, but there can only be maximally 2 lin. indep. eigenvectors.

b. All the solutions of the SDE have the form $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{rt} \begin{pmatrix} a \\ b \end{pmatrix}$ where a and b can be any real numbers, and r is an eigenvalue of the underlying matrix.

If we take $[1;0]$ and $[0;1]$ as the two eigenvectors, it will give general solution $e^{rt} [a; b]$

Star nodes always correspond to scalar matrix $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$

c. The underlying matrix must be a scalar matrix.

d. All the trajectories on the phase portrait start near the origin and move away from the origin. If the star node is stable, the trajectories will start away from origin and move towards the origin.

e. We need to find a generalised eigenvector to form a second solution for the fundamental set of solutions for the SDE. This is only needed if the matrix has only 1 lin.indep. eigenvector.

Long Questions (4 questions; total 45 marks)

Q16 (10 marks)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate Q16 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

The equation of a plane P in the 3D space is given by $ax + by + cz = d$ for some unknown values a , b , c and d . Given that $(1, 1, 3)$ and $(2, -1, -2)$ are two points on the plane.

- (i) [5 marks] Set up a linear system in a , b , c and d , and solve the system. Show your workings clearly.

$$\text{Ans: } \begin{cases} a + b + 3c - d = 0 \\ 2a - b - 2c - d = 0 \end{cases}$$

$$\text{RREF} = \left(\begin{array}{cccc|c} 1 & 0 & 1/3 & -2/3 & 0 \\ 0 & 1 & 8/3 & -1/3 & 0 \end{array} \right), \text{ which gives general solution } \begin{cases} a = \frac{2}{3}t - \frac{1}{3}s \\ b = \frac{1}{3}t - \frac{8}{3}s \\ c = s \\ d = t \end{cases}$$

- (ii) [2 marks] Write down a basis for the solution space of the system in (i).

$$\text{Ans: A basis is given by } \left\{ \begin{pmatrix} 2/3 \\ 1/3 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1/3 \\ -8/3 \\ 1 \\ 0 \end{pmatrix} \right\}$$

- (iii) [3 marks] Write down an equation of the plane P if it contains the origin. Explain how you derive your answer.

Ans: If the plane contains the origin, then $d = 0$. From the general solution, we then

have $a = -\frac{1}{3}s$, $b = -\frac{8}{3}s$, $c = s$, $d = 0$. So an equation of p can be

$$-\frac{1}{3}x - \frac{8}{3}y + z = 0, \text{ or } x + 8y - 3z = 0.$$

Q17 (10 marks)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate Q17 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

There are three quantities x , y , z which are subjected to the following constraints:

$$\begin{cases} x + 2y = 900 \\ y + z = 600 \\ x - 2z = 300 \end{cases}$$

- (i) [2 marks] Show that the above system is inconsistent.

Ans: RREF of augmented matrix is given by $\left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$. Hence inconsistent.

- (ii) [5 marks] Find the least squares solutions of the above system. Show your workings clearly. (You may use MATLAB to compute the standard matrix operations.)

Ans: Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{pmatrix}$, $b = \begin{pmatrix} 900 \\ 600 \\ 300 \end{pmatrix}$. Solve $A^T A x = A^T b$ to get

$$\begin{cases} x = 200 + 2t \\ y = 400 - t \\ z = t \end{cases}$$

- (iii) [3 marks] Suppose the values of x , y , z are all positive whole numbers. Find the particular least squares solution of the system that gives the smallest sum $x + y + z$. Explain how you derive your answer.

Ans: To make sure x , y , z values are positive integers, we need $400 - t > 0$ and $t > 0$.

So the range of t is $0 < t < 400$.

The sum $x + y + z = 600 + 2t$ is minimize when $t = 1$. So the corresponding least squares solution is $x = 202$, $y = 399$, $t = 1$.

Q18 (12 marks)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate Q18 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

In a certain city, there are three video streaming platforms A-flix, B-plus and C-prime. Among them there is a total of 63,000 subscribers (assuming each person only subscribe to one platform at any time). According to market research, for each month, there are movements among the subscribers of the three platforms: 4% and 1% of subscribers of A-flix switch to B-plus and C-prime respective; 6% and 4% of subscribers of B-plus switch to A-flix and C-prime respective; 4% and 4% of subscribers of C-prime switch to A-flix and B-plus respective.

- (i) [3 marks] Let $v_n = \begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix}$, where a_n , b_n and c_n denote the number of subscribers of A-flix, B-plus and C-prime respectively in the n th month. Find the matrix A such that $v_{n+1} = Av_n$.

Ans: $\begin{pmatrix} 0.95 & 0.06 & 0.04 \\ 0.04 & 0.90 & 0.04 \\ 0.01 & 0.04 & 0.92 \end{pmatrix}$.

- (ii) [3 marks] Write down the n -th power A^n of A in the long run. Express the entries of the matrix in fractions. Show your workings clearly. (You may use MATLAB to compute the standard matrix operations.)

Ans: $P = \begin{pmatrix} 32/13 & -1 & 2/3 \\ 18/13 & 0 & -5/3 \\ 1 & 1 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 91/100 & 0 \\ 0 & 0 & 43/50 \end{pmatrix}$.

$A^{\text{large } n} = P \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} P^{-1} = \begin{pmatrix} 32/63 & 32/63 & 32/63 \\ 2/7 & 2/7 & 2/7 \\ 13/63 & 13/63 & 13/63 \end{pmatrix}$

- (iii) [3 marks] What should be the initial number of subscribers a_0 , b_0 and c_0 of the three platforms so that a_n , b_n and c_n will remain unchanged every month? Explain how you derive your answer.

Ans: $v_0 = \begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix}$ must be an eigenvector with eigenvalue 1. i.e. it is a scalar multiple of

$\begin{pmatrix} 32/13 \\ 18/13 \\ 1 \end{pmatrix}$.

Let $\frac{32}{13}k + \frac{18}{13}k + k = 63000$, which gives $k = 13000$.

So $a_0 = 32000$, $b_0 = 18000$, $c_0 = 13000$.

- (iv) [3 marks] Is it possible that, with some initial number of subscribers a_0 , b_0 and c_0 , the three platforms will have equal market share in the long run? Justify your answer.

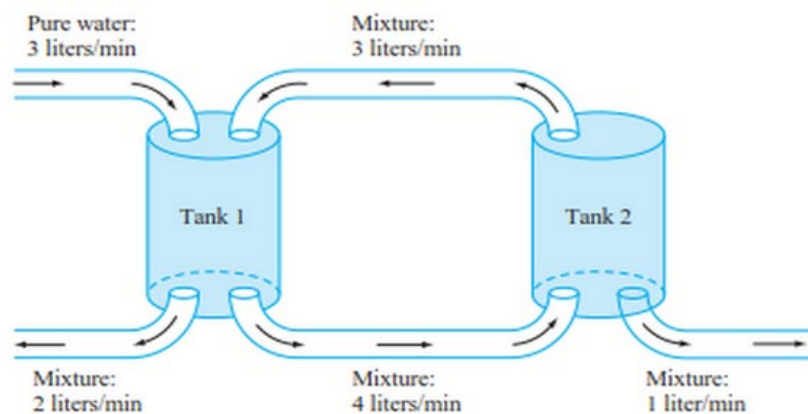
Ans: Not possible. From (ii), regardless of the initial numbers a_0 , b_0 and c_0 , in the long run, the numbers will stabilise as

$\begin{pmatrix} 32/63 & 32/63 & 32/63 \\ 2/7 & 2/7 & 2/7 \\ 13/63 & 13/63 & 13/63 \end{pmatrix} \begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix} = \begin{pmatrix} \frac{32}{63}(a_0 + b_0 + c_0) \\ \frac{2}{7}(a_0 + b_0 + c_0) \\ \frac{13}{63}(a_0 + b_0 + c_0) \end{pmatrix}$

Q19 (13 marks)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate Q19 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

Two tanks are connected by pipes as shown in the diagram. Tank 1 initially contains 20 litres of water in which 50 grams of chlorine are dissolved. Tank 2 initially contains 10 litres of water in which 50 grams of chlorine are dissolved. Beginning at time $t = 0$, pure water is pumped into Tank 1 at a rate of 3 litres per minute, while chlorine/water solutions are exchanged between the tanks and also flow out of both tanks at the rates shown.



- (i) [3 marks] Set up an SDE in terms of the amount of chlorine $y_1(t)$ and $y_2(t)$ in Tank 1 and Tank 2 respectively at time t (in minutes)

Ans: $y_1' = -\frac{6}{20}y_1 + \frac{3}{10}y_2$; $y_2' = \frac{4}{20}y_1 - \frac{4}{10}y_2$

So the underlying matrix of the SDE is $A = \begin{pmatrix} -\frac{6}{20} & \frac{3}{10} \\ \frac{4}{20} & -\frac{4}{10} \end{pmatrix}$

- (ii) [5 marks] Solve the initial value problem (IVP) with the SDE in (i) and the given initial conditions. Show your workings clearly.

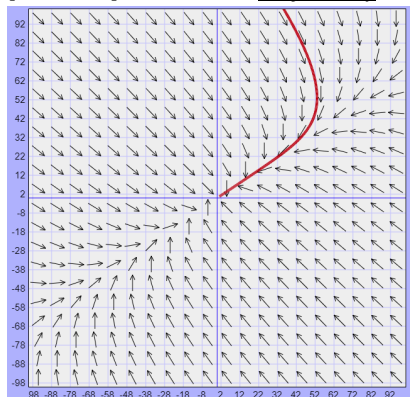
Ans: By MATLAB, the eigenvalues of A are -0.1 and -0.6 , with corresponding eigenvectors

$\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$. So the general solutions are $y = c_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{-t/10} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-6t/10}$.

With the initial condition, we have $3c_1 - c_2 = 50$ and $2c_1 + c_2 = 50$. This gives $c_1 =$

20 , $c_2 = 10$. So the particular solution is $y = 20 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{-t/10} + 10 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-6t/10}$.

- (iii) [3 marks] Sketch the trajectory corresponding to the particular solution of the IVP in (ii).



- (iv) [2 marks] Is it possible to adjust the initial amount of chlorine in the 2 tanks so that there will still be chlorine remains in the tanks in the long run? Justify your answers.

Ans: Not possible. As the two eigenvalues of the underlying matrix are both negative, all trajectories of the phase portrait move close to the origin, which means the chlorine in both tanks will eventually be gone in the long run.

Alternatively, from the general solution $y = c_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{-t/10} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-6t/10}$, we see that as t gets larger, the two exponential functions will tend to 0.